MPG No. 4.1.9 m [Type I, monoclinic] [Q tensor]

* Rank 0 tensor.

$$[Q]$$

$$Q = \mathbb{Q}_0^{(1)}$$

* Rank 1 tensor.

$$\begin{bmatrix} Q_x & 0 & Q_z \end{bmatrix}$$

$$Q_x = \mathbb{Q}_x^{(1)}$$
$$Q_z = \mathbb{Q}_z^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} Q_{xx} & 0 & Q_{xz} \\ 0 & Q_{yy} & 0 \\ Q_{xz} & 0 & Q_{zz} \end{bmatrix}$$

$$Q_{xx} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)}$$
$$Q_{xz} = \mathbb{Q}_{xz}^{(1)}$$
$$Q_{yy} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(1)} - \mathbb{Q}_v^{(1)}$$
$$Q_{zz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_u^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & 0 & Q_{xz} \\ 0 & 0 & 0 \\ -Q_{xz} & 0 & 0 \end{bmatrix}$$

$$Q_{xz} = -\mathbb{G}_y^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} Q_{xxx} & 0 & Q_{xxz} \\ Q_{yyx} & 0 & Q_{yyz} \\ Q_{zzx} & 0 & Q_{zzz} \\ 0 & Q_{yyz} & 0 \\ Q_{xxz} & 0 & Q_{zzx} \\ 0 & Q_{yyx} & 0 \end{bmatrix}$$

$$Q_{xxx} = 3\mathbb{Q}_x^{(1)} + 2\mathbb{Q}_x^{\alpha(1)}$$
$$Q_{xxz} = \mathbb{Q}_z^{(1)} - \mathbb{Q}_z^{\alpha(1)} + \mathbb{Q}_z^{\beta(1)}$$
$$Q_{yyx} = \mathbb{Q}_x^{(1)} - \mathbb{Q}_x^{\alpha(1)} + \mathbb{Q}_x^{\beta(1)}$$
$$Q_{yyz} = \mathbb{Q}_z^{(1)} - \mathbb{Q}_z^{\alpha(1)} - \mathbb{Q}_z^{\beta(1)}$$
$$Q_{zzx} = \mathbb{Q}_x^{(1)} - \mathbb{Q}_x^{\alpha(1)} - \mathbb{Q}_x^{\beta(1)}$$
$$Q_{zzz} = 3\mathbb{Q}_z^{(1)} + 2\mathbb{Q}_z^{\alpha(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & Q_{xxz} \\ Q_{yyx} & 0 & Q_{yyz} \\ Q_{zzx} & 0 & 0 \\ 0 & -\frac{Q_{yyz}}{2} & 0 \\ -\frac{Q_{xxz}}{2} & 0 & -\frac{Q_{zzx}}{2} \\ 0 & -\frac{Q_{yyx}}{2} & 0 \end{bmatrix}$$

$$Q_{xxz} = 2\mathbb{G}^{(1)}_{xy} + 2\mathbb{Q}^{(2)}_{z}$$
$$Q_{yyx} = 2\mathbb{G}^{(1)}_{yz} + 2\mathbb{Q}^{(2)}_{x}$$
$$Q_{yyz} = -2\mathbb{G}^{(1)}_{xy} + 2\mathbb{Q}^{(2)}_{z}$$
$$Q_{zzx} = -2\mathbb{G}^{(1)}_{yz} + 2\mathbb{Q}^{(2)}_{x}$$
$$\frac{Q_{yyz}}{2} = -\mathbb{G}^{(1)}_{xy} + \mathbb{Q}^{(2)}_{z}$$
$$\frac{Q_{xxz}}{2} = \mathbb{G}^{(1)}_{xy} + \mathbb{Q}^{(2)}_{z}$$
$$\frac{Q_{zzx}}{2} = -\mathbb{G}^{(1)}_{yz} + \mathbb{Q}^{(2)}_{x}$$
$$\frac{Q_{yyx}}{2} = \mathbb{G}^{(1)}_{yz} + \mathbb{Q}^{(2)}_{x}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & Q_{yzy} & 0 \\ Q_{zxx} & 0 & Q_{zxz} \\ 0 & Q_{xyy} & 0 \end{bmatrix}$$

$$Q_{yzy} = \mathbb{G}^{(2)}_{xy} + \mathbb{Q}^{(3)}_{z}$$
$$Q_{zxx} = \mathbb{G}^{(2)}_{xy} - \mathbb{Q}^{(3)}_{z}$$
$$Q_{zxz} = \mathbb{G}^{(2)}_{yz} + \mathbb{Q}^{(3)}_{x}$$
$$Q_{xyy} = \mathbb{G}^{(2)}_{yz} - \mathbb{Q}^{(3)}_{x}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & 0 & Q_{xxzx} & 0 \\ Q_{xxyy} & Q_{yyyy} & Q_{yyzz} & 0 & Q_{yyzx} & 0 \\ Q_{xxzz} & Q_{yyzz} & Q_{zzzz} & 0 & Q_{zzzx} & 0 \\ 0 & 0 & 0 & Q_{yzyz} & 0 & Q_{yzxy} \\ Q_{xxzx} & Q_{yyzx} & Q_{zzzx} & 0 & Q_{zxzx} & 0 \\ 0 & 0 & 0 & Q_{yzxy} & 0 & Q_{xyxy} \end{bmatrix}$$

$$Q_{xxxx} = \mathbb{Q}^{(1)}_{0} + 2\mathbb{Q}^{(2)}_{0} - \mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4v} + 2\mathbb{Q}^{(1)}_{4} - 2\mathbb{Q}^{(1)}_{u} - 4\mathbb{Q}^{(2)}_{u} + 2\mathbb{Q}^{(1)}_{v} + 4\mathbb{Q}^{(2)}_{v}$$
$$Q_{xxyy} = \mathbb{Q}^{(1)}_{0} + 2\mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4} - 2\mathbb{Q}^{(1)}_{u}$$
$$Q_{xxzz} = \mathbb{Q}^{(1)}_{0} - \mathbb{Q}^{(1)}_{4u} + \mathbb{Q}^{(1)}_{4v} - \mathbb{Q}^{(1)}_{4} + \mathbb{Q}^{(1)}_{u} + \mathbb{Q}^{(1)}_{v}$$
$$Q_{xxzx} = -\mathbb{Q}^{\alpha(1)}_{4y} - \mathbb{Q}^{\beta(1)}_{4y} + \mathbb{Q}^{(1)}_{xz} + 2\mathbb{Q}^{(2)}_{xz}$$
$$Q_{yyyy} = \mathbb{Q}^{(1)}_{0} + 2\mathbb{Q}^{(2)}_{0} - \mathbb{Q}^{(1)}_{4u} + \mathbb{Q}^{(1)}_{4v} + 2\mathbb{Q}^{(1)}_{4} - 2\mathbb{Q}^{(1)}_{u} - 4\mathbb{Q}^{(2)}_{u} - 2\mathbb{Q}^{(1)}_{v} - 4\mathbb{Q}^{(2)}_{v}$$
$$Q_{yyzz} = \mathbb{Q}^{(1)}_{0} - \mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4v} - \mathbb{Q}^{(1)}_{4} + \mathbb{Q}^{(1)}_{u} - \mathbb{Q}^{(1)}_{v}$$
$$Q_{yyzx} = 2\mathbb{Q}^{\beta(1)}_{4y} + \mathbb{Q}^{(1)}_{xz}$$
$$Q_{zzzz} = \mathbb{Q}^{(1)}_{0} + 2\mathbb{Q}^{(2)}_{0} + 2\mathbb{Q}^{(1)}_{4u} + 2\mathbb{Q}^{(1)}_{4} + 4\mathbb{Q}^{(1)}_{u} + 8\mathbb{Q}^{(2)}_{u}$$
$$Q_{zzzx} = \mathbb{Q}^{\alpha(1)}_{4y} - \mathbb{Q}^{\beta(1)}_{4y} + \mathbb{Q}^{(1)}_{xz} + 2\mathbb{Q}^{(2)}_{xz}$$
$$Q_{yzyz} = \mathbb{Q}^{(2)}_{0} - \mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4v} - \mathbb{Q}^{(1)}_{4} + \mathbb{Q}^{(2)}_{u} - \mathbb{Q}^{(2)}_{v}$$
$$Q_{yzxy} = 2\mathbb{Q}^{\beta(1)}_{4y} + \mathbb{Q}^{(2)}_{xz}$$
$$Q_{zxzx} = \mathbb{Q}^{(2)}_{0} - \mathbb{Q}^{(1)}_{4u} + \mathbb{Q}^{(1)}_{4v} - \mathbb{Q}^{(1)}_{4} + \mathbb{Q}^{(2)}_{u} + \mathbb{Q}^{(2)}_{v}$$
$$Q_{xyxy} = \mathbb{Q}^{(2)}_{0} + 2\mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4} - 2\mathbb{Q}^{(2)}_{u}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & Q_{xxyy} & Q_{xxzz} & 0 & Q_{xxzx} & 0 \\ -Q_{xxyy} & 0 & Q_{yyzz} & 0 & Q_{yyzx} & 0 \\ -Q_{xxzz} & -Q_{yyzz} & 0 & 0 & Q_{zzzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{yzxy} \\ -Q_{xxzx} & -Q_{yyzx} & -Q_{zzzx} & 0 & 0 & 0 \\ 0 & 0 & 0 & -Q_{yzxy} & 0 & 0 \end{bmatrix}$$

$$
\begin{aligned}
Q_{xxyy} &= 4\mathbb{G}^{(1)}_{xyz} - 2\mathbb{Q}^{(3)}_{v} \\
Q_{xxzz} &= -4\mathbb{G}^{(1)}_{xyz} + 3\mathbb{Q}^{(3)}_{u} - \mathbb{Q}^{(3)}_{v} \\
Q_{xxzx} &= -2\mathbb{G}^{(1)}_{y} + 2\mathbb{G}^{\alpha(1)}_{y} + 2\mathbb{G}^{\beta(1)}_{y} + \mathbb{Q}^{(3)}_{xz} \\
Q_{yyzz} &= 4\mathbb{G}^{(1)}_{xyz} + 3\mathbb{Q}^{(3)}_{u} + \mathbb{Q}^{(3)}_{v} \\
Q_{yyzx} &= -4\mathbb{G}^{\beta(1)}_{y} + \mathbb{Q}^{(3)}_{xz} \\
Q_{zzzx} &= 2\mathbb{G}^{(1)}_{y} - 2\mathbb{G}^{\alpha(1)}_{y} + 2\mathbb{G}^{\beta(1)}_{y} + \mathbb{Q}^{(3)}_{xz} \\
Q_{yzxy} &= \mathbb{G}^{(1)}_{y} + 4\mathbb{G}^{\alpha(1)}_{y}
\end{aligned}
$$

* Rank 4 tensor (aas).

$$
\begin{bmatrix}
Q_{yzyz} & 0 & Q_{yzxy} \\
0 & Q_{zxzx} & 0 \\
Q_{yzxy} & 0 & Q_{xyxy}
\end{bmatrix}
$$

$$
\begin{aligned}
Q_{yzyz} &= \mathbb{Q}^{(3)}_{0} - 2\mathbb{Q}^{(4)}_{u} + 2\mathbb{Q}^{(4)}_{v} \\
Q_{yzxy} &= 2\mathbb{Q}^{(4)}_{xz} \\
Q_{zxzx} &= \mathbb{Q}^{(3)}_{0} - 2\mathbb{Q}^{(4)}_{u} - 2\mathbb{Q}^{(4)}_{v} \\
Q_{xyxy} &= \mathbb{Q}^{(3)}_{0} + 4\mathbb{Q}^{(4)}_{u}
\end{aligned}
$$

* Rank 4 tensor (aaa).

$$
\begin{bmatrix}
0 & 0 & Q_{yzxy} \\
0 & 0 & 0 \\
-Q_{yzxy} & 0 & 0
\end{bmatrix}
$$

$$
Q_{yzxy} = \mathbb{G}^{(2)}_{y}
$$

* Rank 4 tensor (sa).

$$
\begin{bmatrix}
0 & Q_{xxzx} & 0 \\
0 & Q_{yyzx} & 0 \\
0 & Q_{zzzx} & 0 \\
Q_{yzyz} & 0 & Q_{yzxy} \\
0 & Q_{zxzx} & 0 \\
Q_{xyyz} & 0 & Q_{xyxy}
\end{bmatrix}
$$

$$
\begin{aligned}
Q_{xxzx} &= \mathbb{G}^{(3)}_{y} - \mathbb{G}^{\alpha(2)}_{y} - \mathbb{G}^{\beta(2)}_{y} + 2\mathbb{Q}^{(5)}_{xz} \\
Q_{yyzx} &= \mathbb{G}^{(3)}_{y} + 2\mathbb{G}^{(4)}_{y} + 2\mathbb{G}^{\alpha(2)}_{y} \\
Q_{zzzx} &= \mathbb{G}^{(3)}_{y} - \mathbb{G}^{\alpha(2)}_{y} + \mathbb{G}^{\beta(2)}_{y} - 2\mathbb{Q}^{(5)}_{xz} \\
Q_{yzyz} &= \mathbb{G}^{(2)}_{xyz} - 3\mathbb{Q}^{(5)}_{u} - \mathbb{Q}^{(5)}_{v} \\
Q_{yzxy} &= \mathbb{G}^{(4)}_{y} - \mathbb{G}^{\alpha(2)}_{y} + \mathbb{G}^{\beta(2)}_{y} + \mathbb{Q}^{(5)}_{xz} \\
Q_{zxzx} &= \mathbb{G}^{(2)}_{xyz} + 3\mathbb{Q}^{(5)}_{u} - \mathbb{Q}^{(5)}_{v} \\
Q_{xyyz} &= \mathbb{G}^{(4)}_{y} - \mathbb{G}^{\alpha(2)}_{y} - \mathbb{G}^{\beta(2)}_{y} - \mathbb{Q}^{(5)}_{xz} \\
Q_{xyxy} &= \mathbb{G}^{(2)}_{xyz} + 2\mathbb{Q}^{(5)}_{v}
\end{aligned}
$$

* Rank 4 tensor (as).

$$
\begin{bmatrix}
0 & 0 & 0 & Q_{yzyz} & 0 & Q_{yzxy} \\
Q_{zxxx} & Q_{zxyy} & Q_{zxzz} & 0 & Q_{zxzx} & 0 \\
0 & 0 & 0 & Q_{xyyz} & 0 & Q_{xyxy}
\end{bmatrix}
$$

$$
\begin{aligned}
Q_{yzyz} &= \mathbb{G}^{(3)}_{xyz} - 3\mathbb{Q}^{(6)}_{u} - \mathbb{Q}^{(6)}_{v} \\
Q_{yzxy} &= \mathbb{G}^{(6)}_{y} - \mathbb{G}^{\alpha(3)}_{y} - \mathbb{G}^{\beta(3)}_{y} - \mathbb{Q}^{(6)}_{xz} \\
Q_{zxxx} &= \mathbb{G}^{(5)}_{y} - \mathbb{G}^{\alpha(3)}_{y} - \mathbb{G}^{\beta(3)}_{y} + 2\mathbb{Q}^{(6)}_{xz} \\
Q_{zxyy} &= \mathbb{G}^{(5)}_{y} + 2\mathbb{G}^{(6)}_{y} + 2\mathbb{G}^{\alpha(3)}_{y} \\
Q_{zxzz} &= \mathbb{G}^{(5)}_{y} - \mathbb{G}^{\alpha(3)}_{y} + \mathbb{G}^{\beta(3)}_{y} - 2\mathbb{Q}^{(6)}_{xz} \\
Q_{zxzx} &= \mathbb{G}^{(3)}_{xyz} + 3\mathbb{Q}^{(6)}_{u} - \mathbb{Q}^{(6)}_{v} \\
Q_{xyyz} &= \mathbb{G}^{(6)}_{y} - \mathbb{G}^{\alpha(3)}_{y} + \mathbb{G}^{\beta(3)}_{y} + \mathbb{Q}^{(6)}_{xz} \\
Q_{xyxy} &= \mathbb{G}^{(3)}_{xyz} + 2\mathbb{Q}^{(6)}_{v}
\end{aligned}
$$

* Rank 4 tensor (s).

$$
\begin{bmatrix}
Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & 0 & Q_{xxzx} & 0 & 0 & Q_{xxxz} & 0 \\
Q_{yyxx} & Q_{yyyy} & Q_{yyzz} & 0 & Q_{yyzx} & 0 & 0 & Q_{yyxz} & 0 \\
Q_{zzxx} & Q_{zzyy} & Q_{zzzz} & 0 & Q_{zzzx} & 0 & 0 & Q_{zzxz} & 0 \\
0 & 0 & 0 & Q_{yzyz} & 0 & Q_{yzxy} & Q_{yzzy} & 0 & Q_{yzyx} \\
Q_{zxxx} & Q_{zxyy} & Q_{zxzz} & 0 & Q_{zxzx} & 0 & 0 & Q_{zxxz} & 0 \\
0 & 0 & 0 & Q_{xyyz} & 0 & Q_{xyxy} & Q_{xyzy} & 0 & Q_{xyyx}
\end{bmatrix}
$$

$$Q_{xxxx} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} + 2\mathbb{Q}_v^{(1)} + 4\mathbb{Q}_v^{(2)}$$
$$Q_{xxyy} = 4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 2\mathbb{Q}_v^{(3)}$$
$$Q_{xxzz} = -4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)} + \mathbb{Q}_v^{(1)} - \mathbb{Q}_v^{(3)}$$
$$Q_{xzxx} = -2\mathbb{G}_y^{(1)} + \mathbb{G}_y^{(3)} + 2\mathbb{G}_y^{\alpha(1)} - \mathbb{G}_y^{\alpha(2)} + 2\mathbb{G}_y^{\beta(1)} - \mathbb{G}_y^{\beta(2)} - \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} + 2\mathbb{Q}_{xz}^{(5)}$$
$$Q_{xxzx} = -2\mathbb{G}_y^{(1)} - \mathbb{G}_y^{(3)} + 2\mathbb{G}_y^{\alpha(1)} + \mathbb{G}_y^{\alpha(2)} + 2\mathbb{G}_y^{\beta(1)} + \mathbb{G}_y^{\beta(2)} - \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} - 2\mathbb{Q}_{xz}^{(5)}$$
$$Q_{yyxx} = -4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} + 2\mathbb{Q}_v^{(3)}$$
$$Q_{yyyy} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} - 2\mathbb{Q}_v^{(1)} - 4\mathbb{Q}_v^{(2)}$$
$$Q_{yyzz} = 4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)} - \mathbb{Q}_v^{(1)} + \mathbb{Q}_v^{(3)}$$
$$Q_{yyzx} = \mathbb{G}_y^{(3)} + 2\mathbb{G}_y^{(4)} + 2\mathbb{G}_y^{\alpha(2)} - 4\mathbb{G}_y^{\beta(1)} + 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + \mathbb{Q}_{xz}^{(3)}$$
$$Q_{yyxz} = -\mathbb{G}_y^{(3)} - 2\mathbb{G}_y^{(4)} - 2\mathbb{G}_y^{\alpha(2)} - 4\mathbb{G}_y^{\beta(1)} + 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + \mathbb{Q}_{xz}^{(3)}$$
$$Q_{zzxx} = 4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)} + \mathbb{Q}_v^{(1)} + \mathbb{Q}_v^{(3)}$$
$$Q_{zzyy} = -4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)} - \mathbb{Q}_v^{(1)} - \mathbb{Q}_v^{(3)}$$
$$Q_{zzzz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)}$$
$$Q_{zzzx} = 2\mathbb{G}_y^{(1)} + \mathbb{G}_y^{(3)} - 2\mathbb{G}_y^{\alpha(1)} - \mathbb{G}_y^{\alpha(2)} + 2\mathbb{G}_y^{\beta(1)} + \mathbb{G}_y^{\beta(2)} + \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} - 2\mathbb{Q}_{xz}^{(5)}$$
$$Q_{zzxz} = 2\mathbb{G}_y^{(1)} - \mathbb{G}_y^{(3)} - 2\mathbb{G}_y^{\alpha(1)} + \mathbb{G}_y^{\alpha(2)} + 2\mathbb{G}_y^{\beta(1)} - \mathbb{G}_y^{\beta(2)} + \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} + 2\mathbb{Q}_{xz}^{(5)}$$
$$Q_{yzyz} = \mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} - 3\mathbb{Q}_u^{(5)} - \mathbb{Q}_v^{(2)} - \mathbb{Q}_v^{(5)}$$
$$Q_{yzxy} = \mathbb{G}_y^{(1)} + \mathbb{G}_y^{(4)} + 4\mathbb{G}_y^{\alpha(1)} - \mathbb{G}_y^{\alpha(2)} + \mathbb{G}_y^{\beta(2)} + 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(5)}$$
$$Q_{yzzy} = -\mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)} - \mathbb{Q}_v^{(2)} + \mathbb{Q}_v^{(5)}$$
$$Q_{yzyx} = \mathbb{G}_y^{(1)} - \mathbb{G}_y^{(4)} + 4\mathbb{G}_y^{\alpha(1)} + \mathbb{G}_y^{\alpha(2)} - \mathbb{G}_y^{\beta(2)} + 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(2)} - \mathbb{Q}_{xz}^{(5)}$$
$$Q_{zxxx} = 2\mathbb{G}_y^{(1)} - 2\mathbb{G}_y^{\alpha(1)} - 2\mathbb{G}_y^{\beta(1)} - \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} - \mathbb{Q}_{xz}^{(3)}$$
$$Q_{zxyy} = 4\mathbb{G}_y^{\beta(1)} + 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} - \mathbb{Q}_{xz}^{(3)}$$
$$Q_{zxzz} = -2\mathbb{G}_y^{(1)} + 2\mathbb{G}_y^{\alpha(1)} - 2\mathbb{G}_y^{\beta(1)} + \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} - \mathbb{Q}_{xz}^{(3)}$$
$$Q_{zxzx} = \mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)} + \mathbb{Q}_v^{(2)} - \mathbb{Q}_v^{(5)}$$
$$Q_{zxxz} = -\mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} - 3\mathbb{Q}_u^{(5)} + \mathbb{Q}_v^{(2)} + \mathbb{Q}_v^{(5)}$$
$$Q_{xyyz} = -\mathbb{G}_y^{(1)} + \mathbb{G}_y^{(4)} - 4\mathbb{G}_y^{\alpha(1)} - \mathbb{G}_y^{\alpha(2)} - \mathbb{G}_y^{\beta(2)} + 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(2)} - \mathbb{Q}_{xz}^{(5)}$$
$$Q_{xyxy} = \mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(2)} + 2\mathbb{Q}_v^{(5)}$$
$$Q_{xyzy} = -\mathbb{G}_y^{(1)} - \mathbb{G}_y^{(4)} - 4\mathbb{G}_y^{\alpha(1)} + \mathbb{G}_y^{\alpha(2)} + \mathbb{G}_y^{\beta(2)} + 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(5)}$$
$$Q_{xyyx} = -\mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(2)} - 2\mathbb{Q}_v^{(5)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & Q_{yzyz} & 0 & Q_{yzxy} & Q_{yzzy} & 0 & Q_{yzyx} \\ Q_{zxxx} & Q_{zxyy} & Q_{zxzz} & 0 & Q_{zxzx} & 0 & 0 & Q_{zxxz} & 0 \\ 0 & 0 & 0 & Q_{xyyz} & 0 & Q_{xyxy} & Q_{xyzy} & 0 & Q_{xyyx} \end{bmatrix}$$

$$
\begin{aligned}
Q_{yzyz} &= \mathbb{G}^{(3)}_{xyz} + \mathbb{Q}^{(3)}_{0} - 2\mathbb{Q}^{(4)}_{u} - 3\mathbb{Q}^{(6)}_{u} + 2\mathbb{Q}^{(4)}_{v} - \mathbb{Q}^{(6)}_{v} \\
Q_{yzxy} &= \mathbb{G}^{(2)}_{y} + \mathbb{G}^{(6)}_{y} - \mathbb{G}^{\alpha(3)}_{y} - \mathbb{G}^{\beta(3)}_{y} + 2\mathbb{Q}^{(4)}_{xz} - \mathbb{Q}^{(6)}_{xz} \\
Q_{yzzy} &= \mathbb{G}^{(3)}_{xyz} - \mathbb{Q}^{(3)}_{0} + 2\mathbb{Q}^{(4)}_{u} - 3\mathbb{Q}^{(6)}_{u} - 2\mathbb{Q}^{(4)}_{v} - \mathbb{Q}^{(6)}_{v} \\
Q_{yzyx} &= -\mathbb{G}^{(2)}_{y} + \mathbb{G}^{(6)}_{y} - \mathbb{G}^{\alpha(3)}_{y} - \mathbb{G}^{\beta(3)}_{y} - 2\mathbb{Q}^{(4)}_{xz} - \mathbb{Q}^{(6)}_{xz} \\
Q_{zxxx} &= \mathbb{G}^{(5)}_{y} - \mathbb{G}^{\alpha(3)}_{y} - \mathbb{G}^{\beta(3)}_{y} + 2\mathbb{Q}^{(6)}_{xz} \\
Q_{zxyy} &= \mathbb{G}^{(5)}_{y} + 2\mathbb{G}^{(6)}_{y} + 2\mathbb{G}^{\alpha(3)}_{y} \\
Q_{zxzz} &= \mathbb{G}^{(5)}_{y} - \mathbb{G}^{\alpha(3)}_{y} + \mathbb{G}^{\beta(3)}_{y} - 2\mathbb{Q}^{(6)}_{xz} \\
Q_{zxzx} &= \mathbb{G}^{(3)}_{xyz} + \mathbb{Q}^{(3)}_{0} - 2\mathbb{Q}^{(4)}_{u} + 3\mathbb{Q}^{(6)}_{u} - 2\mathbb{Q}^{(4)}_{v} - \mathbb{Q}^{(6)}_{v} \\
Q_{zxxz} &= \mathbb{G}^{(3)}_{xyz} - \mathbb{Q}^{(3)}_{0} + 2\mathbb{Q}^{(4)}_{u} + 3\mathbb{Q}^{(6)}_{u} + 2\mathbb{Q}^{(4)}_{v} - \mathbb{Q}^{(6)}_{v} \\
Q_{xyyz} &= -\mathbb{G}^{(2)}_{y} + \mathbb{G}^{(6)}_{y} - \mathbb{G}^{\alpha(3)}_{y} + \mathbb{G}^{\beta(3)}_{y} + 2\mathbb{Q}^{(4)}_{xz} + \mathbb{Q}^{(6)}_{xz} \\
Q_{xyxy} &= \mathbb{G}^{(3)}_{xyz} + \mathbb{Q}^{(3)}_{0} + 4\mathbb{Q}^{(4)}_{u} + 2\mathbb{Q}^{(6)}_{v} \\
Q_{xyzy} &= \mathbb{G}^{(2)}_{y} + \mathbb{G}^{(6)}_{y} - \mathbb{G}^{\alpha(3)}_{y} + \mathbb{G}^{\beta(3)}_{y} - 2\mathbb{Q}^{(4)}_{xz} + \mathbb{Q}^{(6)}_{xz} \\
Q_{xyyx} &= \mathbb{G}^{(3)}_{xyz} - \mathbb{Q}^{(3)}_{0} - 4\mathbb{Q}^{(4)}_{u} + 2\mathbb{Q}^{(6)}_{v}
\end{aligned}
$$

* Rank 4 tensor (t).

$$
\begin{bmatrix}
Q_{xxxx} & 0 & Q_{xxxz} \\
0 & Q_{yyyy} & 0 \\
Q_{zzzx} & 0 & Q_{zzzz} \\
Q_{yyzx} & 0 & Q_{yyzz} \\
Q_{zzxx} & 0 & Q_{zzxz} \\
0 & Q_{xxyy} & 0 \\
0 & Q_{yzzy} & 0 \\
Q_{zxxx} & 0 & Q_{zxxz} \\
Q_{xxyy} + Q_{yyzz} - Q_{yzzy} - Q_{zxxz} + Q_{zzxx} & 0 & Q_{xyyz} \\
0 & Q_{xyyz} + \frac{Q_{zzxz}}{2} - \frac{Q_{zzzx}}{2} & 0
\end{bmatrix}
$$

$$
\begin{aligned}
Q_{xxxx} &= 3\mathbb{Q}^{(1)}_{0} - \mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4v} + 2\mathbb{Q}^{(1)}_{4} - 2\mathbb{Q}^{(1)}_{u} - 4\mathbb{Q}^{(2)}_{u} + 2\mathbb{Q}^{(1)}_{v} + 4\mathbb{Q}^{(2)}_{v} \\
Q_{xxxz} &= -\mathbb{Q}^{\alpha(1)}_{4y} - \mathbb{Q}^{\beta(1)}_{4y} + \mathbb{Q}^{(1)}_{xz} + 2\mathbb{Q}^{(2)}_{xz} + \mathbb{Q}^{(3)}_{xz} - 2\mathbb{Q}^{(5)}_{xz} \\
Q_{yyyy} &= 3\mathbb{Q}^{(1)}_{0} - \mathbb{Q}^{(1)}_{4u} + \mathbb{Q}^{(1)}_{4v} + 2\mathbb{Q}^{(1)}_{4} - 2\mathbb{Q}^{(1)}_{u} - 4\mathbb{Q}^{(2)}_{u} - 2\mathbb{Q}^{(1)}_{v} - 4\mathbb{Q}^{(2)}_{v} \\
Q_{zzzx} &= \mathbb{Q}^{\alpha(1)}_{4y} - \mathbb{Q}^{\beta(1)}_{4y} + \mathbb{Q}^{(1)}_{xz} + 2\mathbb{Q}^{(2)}_{xz} + \mathbb{Q}^{(3)}_{xz} - 2\mathbb{Q}^{(5)}_{xz} \\
Q_{zzzz} &= 3\mathbb{Q}^{(1)}_{0} + 2\mathbb{Q}^{(1)}_{4u} + 2\mathbb{Q}^{(1)}_{4} + 4\mathbb{Q}^{(1)}_{u} + 8\mathbb{Q}^{(2)}_{u} \\
Q_{yyzx} &= 2\mathbb{Q}^{\beta(1)}_{4y} + \mathbb{Q}^{(1)}_{xz} + \mathbb{Q}^{(3)}_{xz} \\
Q_{yyzz} &= \mathbb{Q}^{(1)}_{0} - \mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4v} - \mathbb{Q}^{(1)}_{4} + \mathbb{Q}^{(1)}_{u} + 3\mathbb{Q}^{(3)}_{u} - \mathbb{Q}^{(1)}_{v} + \mathbb{Q}^{(3)}_{v} \\
Q_{zzxx} &= \mathbb{Q}^{(1)}_{0} - \mathbb{Q}^{(1)}_{4u} + \mathbb{Q}^{(1)}_{4v} - \mathbb{Q}^{(1)}_{4} + \mathbb{Q}^{(1)}_{u} - 3\mathbb{Q}^{(3)}_{u} + \mathbb{Q}^{(1)}_{v} + \mathbb{Q}^{(3)}_{v} \\
Q_{zzxz} &= \mathbb{Q}^{\alpha(1)}_{4y} - \mathbb{Q}^{\beta(1)}_{4y} + \mathbb{Q}^{(1)}_{xz} + 2\mathbb{Q}^{(2)}_{xz} + \mathbb{Q}^{(3)}_{xz} + 2\mathbb{Q}^{(5)}_{xz} \\
Q_{xxyy} &= \mathbb{Q}^{(1)}_{0} + 2\mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4} - 2\mathbb{Q}^{(1)}_{u} - 2\mathbb{Q}^{(3)}_{v} \\
Q_{yzzy} &= \mathbb{Q}^{(1)}_{0} - \mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4v} - \mathbb{Q}^{(1)}_{4} + \mathbb{Q}^{(2)}_{u} + 3\mathbb{Q}^{(5)}_{u} - \mathbb{Q}^{(2)}_{v} + \mathbb{Q}^{(5)}_{v} \\
Q_{zxxx} &= -\mathbb{Q}^{\alpha(1)}_{4y} - \mathbb{Q}^{\beta(1)}_{4y} + \mathbb{Q}^{(1)}_{xz} + 2\mathbb{Q}^{(2)}_{xz} - \mathbb{Q}^{(3)}_{xz} \\
Q_{zxxz} &= \mathbb{Q}^{(1)}_{0} - \mathbb{Q}^{(1)}_{4u} + \mathbb{Q}^{(1)}_{4v} - \mathbb{Q}^{(1)}_{4} + \mathbb{Q}^{(2)}_{u} - 3\mathbb{Q}^{(5)}_{u} + \mathbb{Q}^{(2)}_{v} + \mathbb{Q}^{(5)}_{v} \\
Q_{xyyz} &= 2\mathbb{Q}^{\beta(1)}_{4y} + \mathbb{Q}^{(2)}_{xz} - \mathbb{Q}^{(5)}_{xz}
\end{aligned}
$$